Online Appendix
"Schools as Safety-nets: Break-downs and Recovery in Reporting of Violence Against Children"

Pilar Larroulet, Daniel Pailañir, Daniela Quintana & Damian Clarke

A1 Supplementary Figures

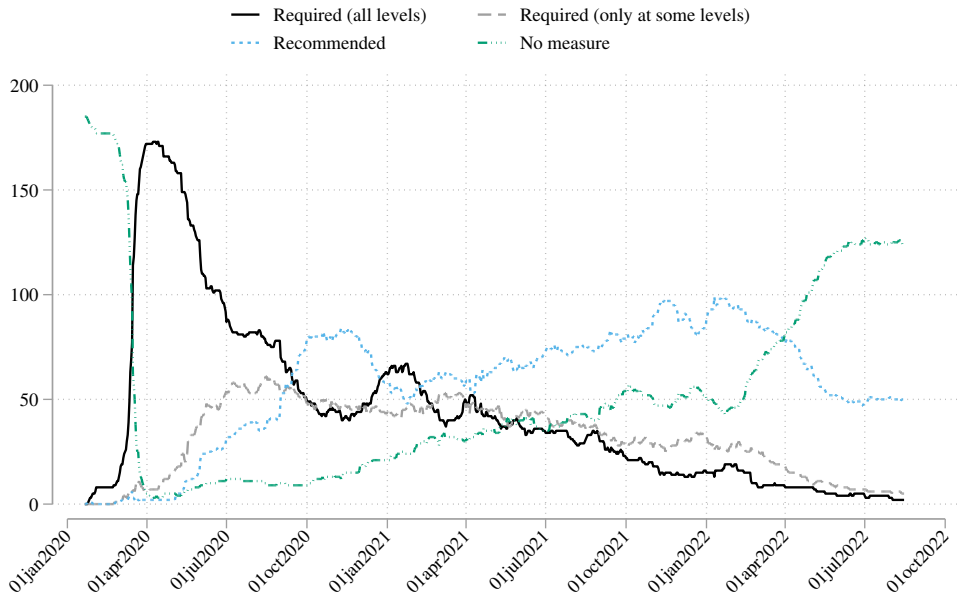


Figure A1: Country-Level School Closure Policies

Country level school closure policies by time are plotted, as classified by Hale et al. (2021).

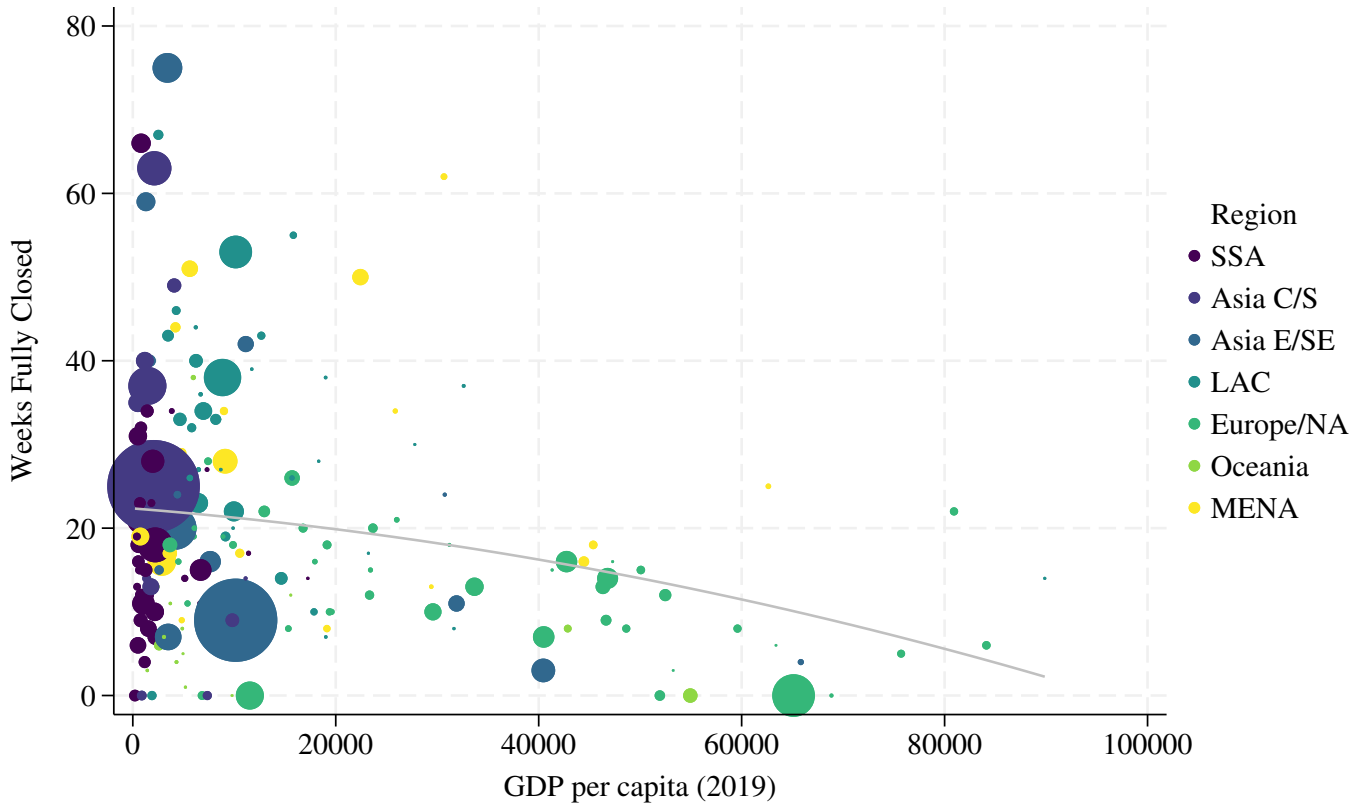


Figure A2: Duration of School Closures and Country Income Levels

Notes to Figure A2: Scatter plot displays the length of school closures as reported by UNESCO (2022), and GDP per capita in 2019 (pre-school closures). Each point represents a single country, and the size of each point indicates the total enrollment for this country. Points are colored by region. The gray line plots a quadratic fit. A very small number of countries with GDP above 100,000 USD per capita have been omitted for ease of vizualisation.

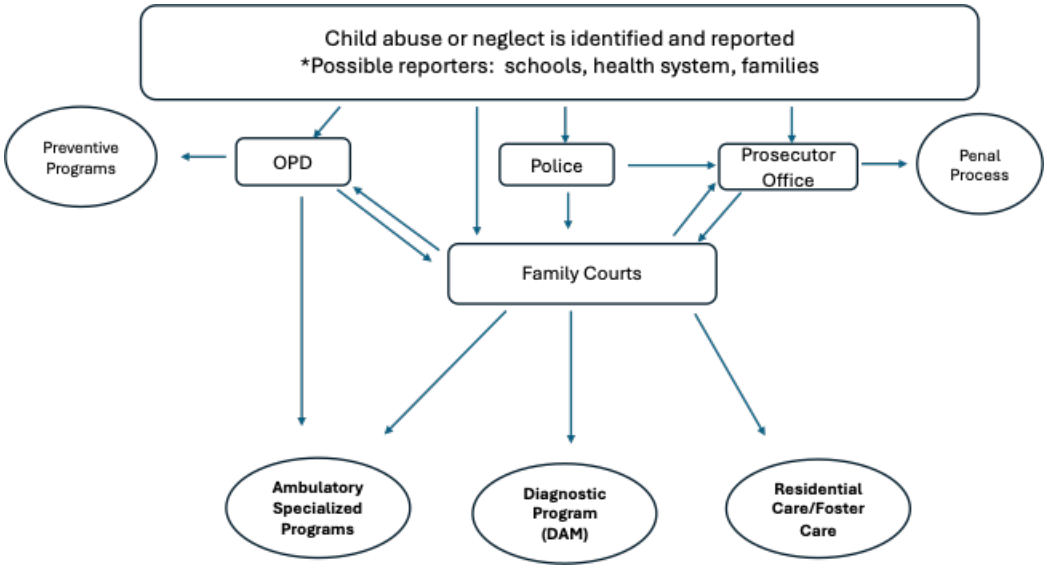
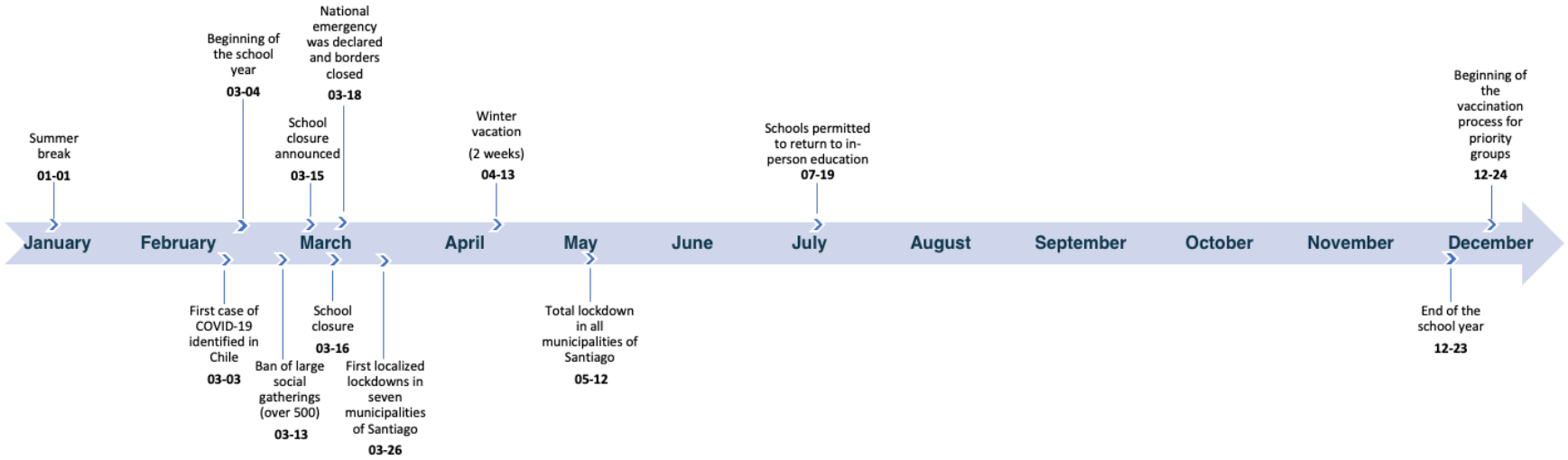
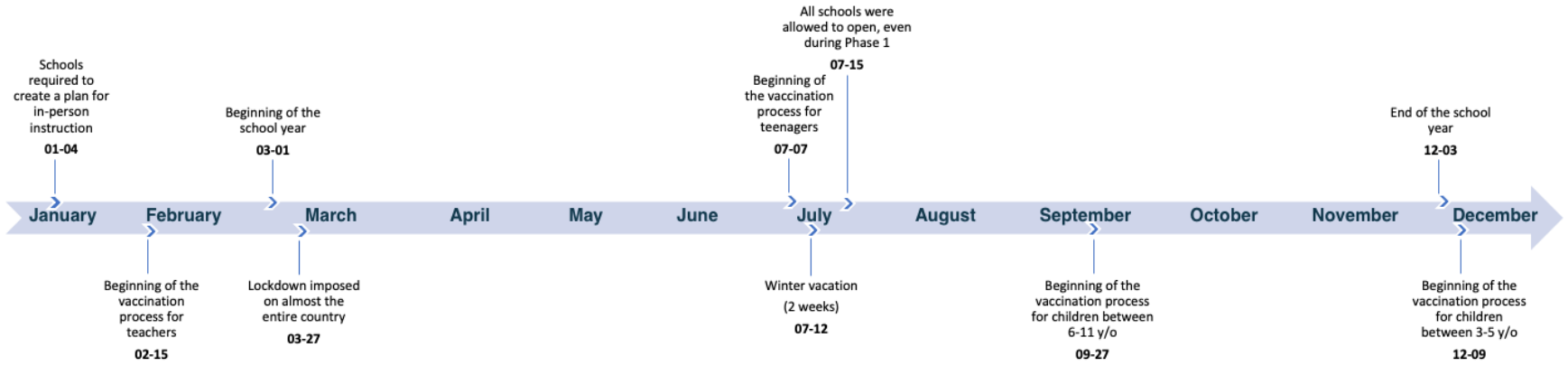


Figure A3: Child Protection Procedures in Chile

Notes to Figure A3: Figure based on information from de Iruarrizaga (2016); SENAME (2016); Stutzin Vallejos (2018). In **bold**, different types of programs under the supervision of SENAME/Mejor Niñez. OPDs are the local offices responsible for providing support and protection to children and adolescents who have experienced rights violations. Co-funded and overseen by SENAME, these offices have two main functions: delivering intervention programs for less complex cases and referring higher-risk cases to Family Courts and other programs; and coordinating inter-sectoral efforts to develop the local protection system. As part of their mission, OPDs are tasked with promoting the rights of children and adolescents and developing municipal policies for their protection. They are currently in the process of being replaced by the Local Children's Offices. Procedures reflect how the system worked until October 2021.



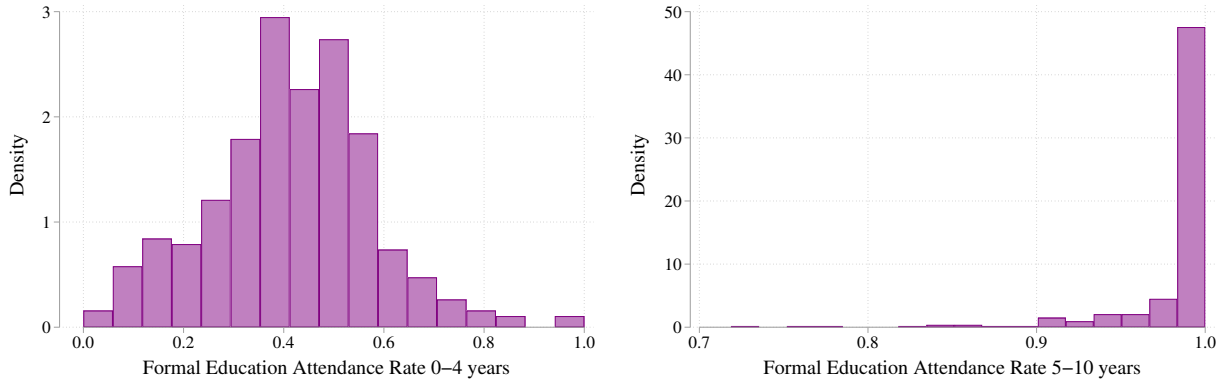
(a) Year 2020



(b) Year 2021

Figure A4: Key Events: 2020–2021

Notes to Figure A4: Key dates are plotted from January-December 2020 (top panel) and January-December 2021 (bottom panel). Precise dates (MM-DD) are noted below events in text.



(a) Ages 0–4

(b) Ages 5–10

Figure A5: Attendance Rate by Age

Notes to Figure A5: Each panel presents attendance rates estimated at the municipality level among children aged 0-4 (panel (a)) and 5-10 (panel (b)). Rates are estimated as the proportion of children attending in each municipality observed in a large nationally representative national survey (the CASEN) in the pre-pandemic period, with each observation referring to a single municipality.

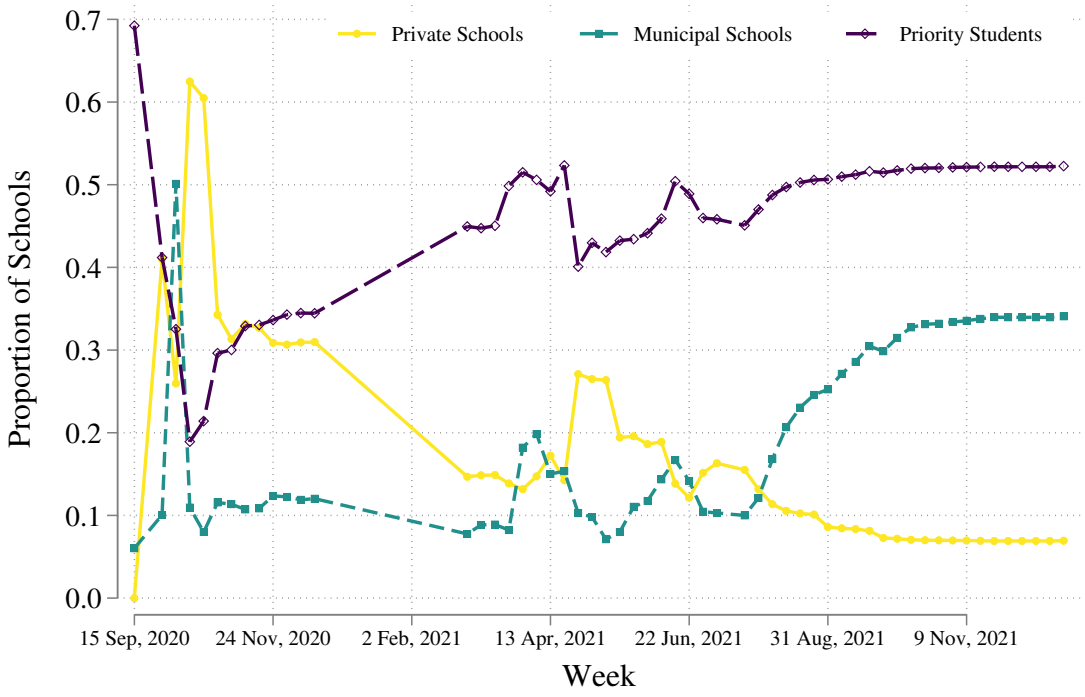
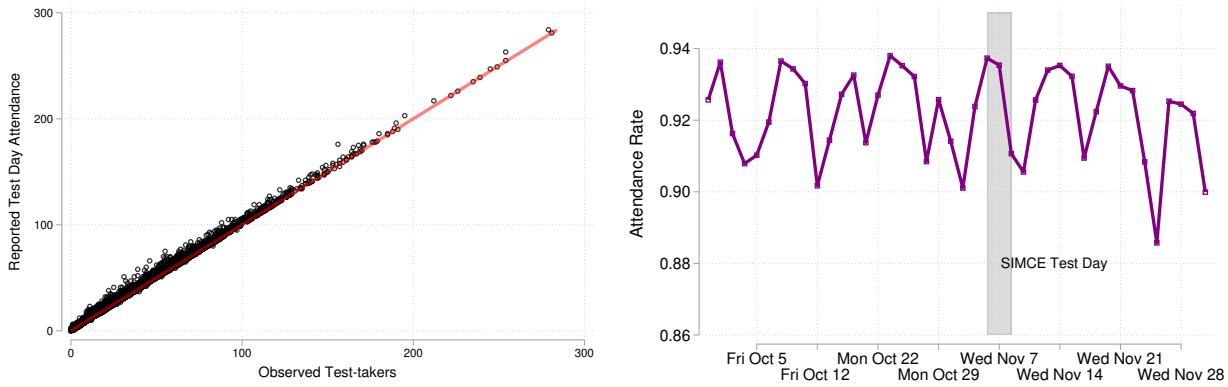


Figure A6: Characteristics of Open Schools

Notes to Figure A6: Each line plots the proportion of open schools which are private schools or municipal-run (public schools), as well as the proportion of students classified as priority among open schools. This, trends capture the change in compositions of open schools as school reopening occurs from September 2020 (first reopenings), up until November 2021 (all schools reopened).



(a) Test Day to Administrative Correspondence

(b) Attendance Surrounding Test Day

Figure A7: Administrate Attendance Records Consistency Check

Notes to Figure A7: Left-hand panel documents reported attendance on the day of the fourth grade SIMCE mathematics test in 2018 (y-axis) versus actual observed number of students completing tests. Each point is a single school in the country. The 45 degree red line indicates a 1:1 correspondence between the quantities on each axis. Right-hand panel documents attendance rates among all fourth grade students on all school days around the SIMCE test, where the gray shaded area (November 7th) indicates the test day for mathematics. Note that in the right-hand plot, some weeks have fewer than 5 days given public holidays in early November and mid-October.



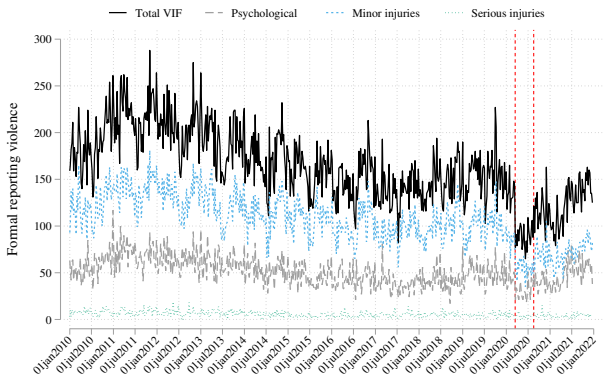
Figure A8: Administrative Records of School Reopening Across Chile

Notes to Figure A8: School opening is displayed across the entire of country of Chile for weeks beginning 5 August 2020 (the week of first reopening is 17 August 2020), up to 27 September 2021. After this date, schools were nearly entirely reopened (see Figure 1). Proportion of schools reopened are displayed at the regional level for each of Chile's 16 regions, and refers to the proportion of all students whose school is reopened based on administrative records of student enrollment, and school reopenings.

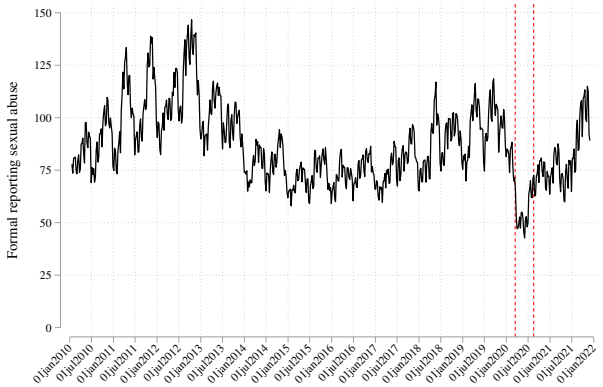


Figure A9: Administrative Records of School Reopening Across Chile's Metropolitan Region

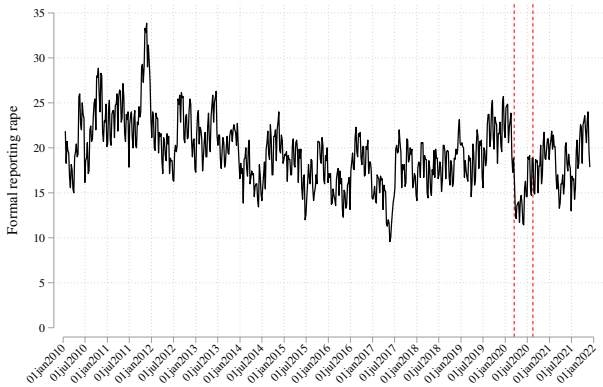
Notes to Figure A9: School opening is displayed across the Metropolitan Region of Santiago (the capital of Chile) for weeks beginning 5 August 2020 (the week of first reopening is 17 August 2020), and ending 27 September 2021. After this date, schools were nearly entirely reopened (see Figure 1). Proportion of schools reopened are displayed at the municipal level for each of the Metropolitan Region's 32 municipalities (of the total of 346 municipalities in the country), and refers to the proportion of all students whose school is reopened based on administrative records of student enrollment, and school reopenings.



(a) Intra-family Violence



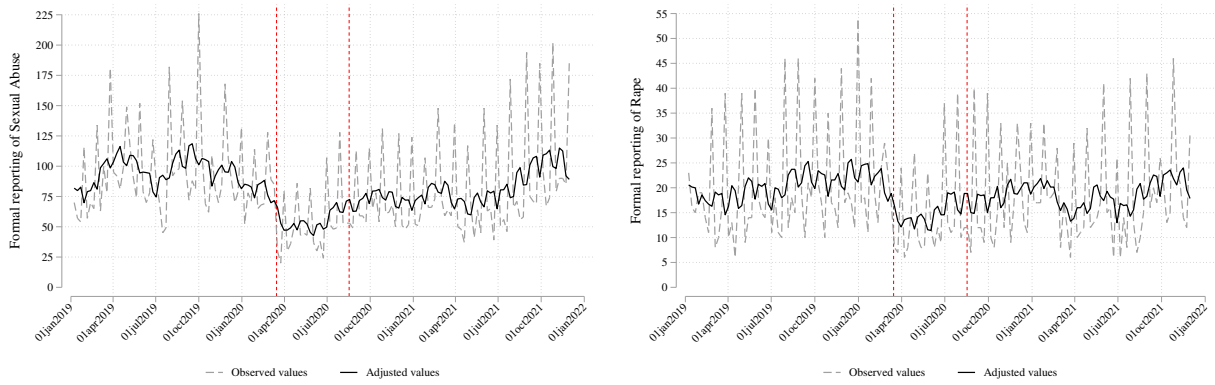
(b) Sexual Abuse



(c) Rape

Figure A10: Extended Trends: Intra-family violence, Sexual Abuse and Rape Against Minors

Notes to Figure A10: Trends show the weekly number of formal complaints received by police related to intra-family violence (panel (a)), sexual abuse (panel (b)), and rape (panel (c)) against individuals aged under 18 years. Here, longer trends in outcomes are documented, dating from January 1, 2010 to 31 December, 2021. In main analysis, the period of January 1, 2019 to 31 December, 2021 is used. Vertical red lines denote school closures and the date of first re-opening. Panel (a) additionally breaks down total intra-family violence (black solid line), into complaints classified as psychological violence, minor injuries, or serious injuries.

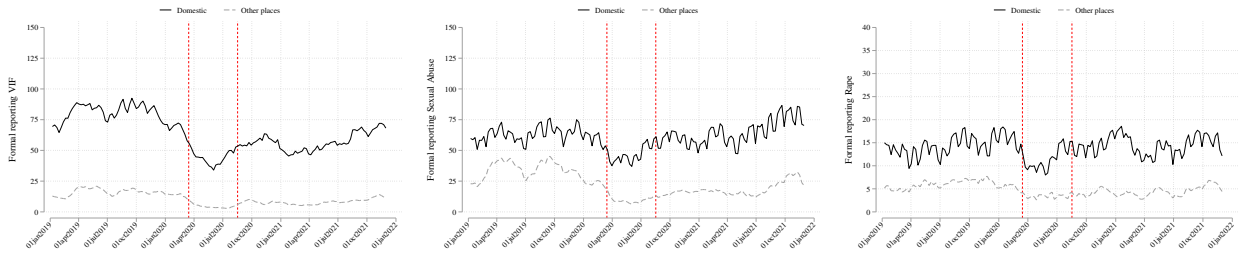


(a) Reporting of sexual assault against minors

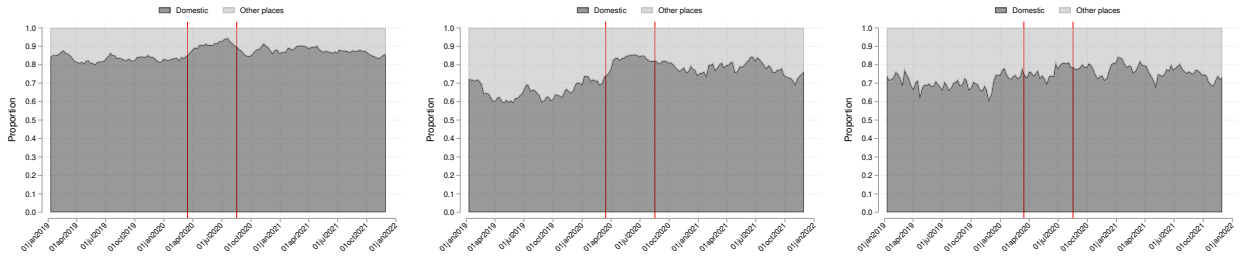
(b) Reporting of rape against minors

Figure A11: Temporal Trends – Sexual abuse and Rape against Minors, Smoothed and Unsmoothed Outcomes

Notes to Figure A11: Trends show the total number of cases of Sexual assault (panel (a)), and rape (panel (b)) against minors according to original records which tend to over-assign dates as the first day of each month, and smoothed records where these over assigned cases have been uniformly reassigned in each municipality within each month. In principal analysis smoothed values (solid black line) are used, as these are closer to the actual occurrence of crimes. In Appendix Results we document results using original unsmoothed measures.



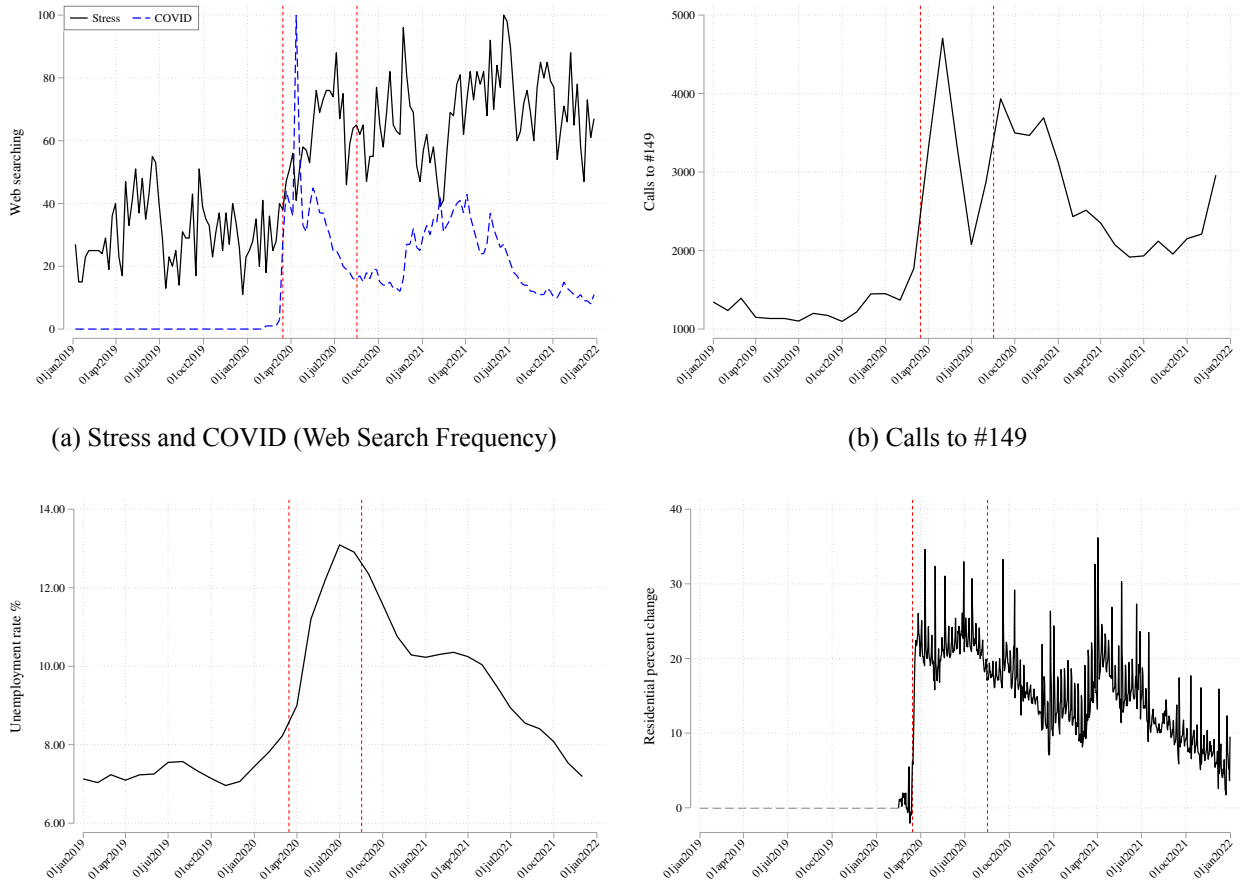
(a) Reporting of Intra-family Violence Against Minors (b) Reporing of Sexual Assault Against Minors (c) Reporting of Rape Against Minors



(d) Proportion of Reporting Violence Against Minors (e) Proportion of Reporting of Sexual Assault Against Minors (f) Proportion of Reporting of Rape Against Minors

Figure A12: Temporal Trends – Crimes Reported Against Children by Place of Occurrence

Notes to Figure A12: Descriptive trends show the total number of crimes against minors (top panels) and proportion of crimes against minors (bottom panels) classified as occurring in a private home ("Domestic"), or in another place ("Other places"). The total number and proportion of cases are documented by week for the full period of analysis. Information on the location of occurrence is not available in data on crime victims, but rather an auxiliary database covering all crimes. As crimes can have more than 1 victim, these descriptive trends have slightly less crimes than victimization data documented in Figure 1 of the main analysis. In all cases, crimes are defined as in the main analysis (intra-family violence in panels (a) and (d), sexual assault in panels (b) and (e), and rape in panels (c) and (f)). Vertical red lines document the data of first school closure, and first reopening.



(a) Stress and COVID (Web Search Frequency)

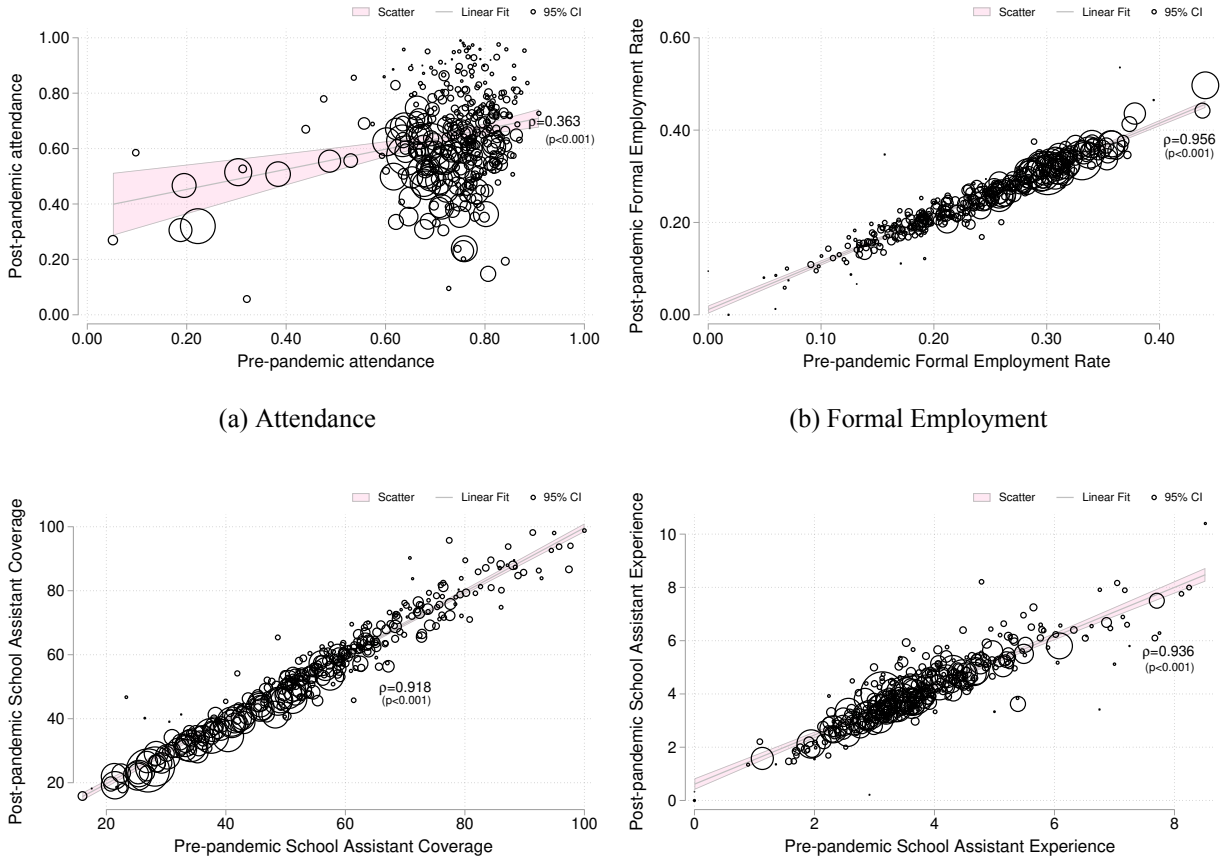
(b) Calls to #149

(c) Unemployment

(d) Home time

Figure A13: Trends in Other Relevant Factors

Notes to Figure A13: Trends by day (panel (d)), week (panel (a)), or month (panels (b) and (c)) of other relevant factors in Chile are displayed over time. Panel (a) documents relative frequency of online search based on Google trends data in Chile for the term 'stress' (*estrés*) and COVID (for comparison), suggesting increases in searches related to stress following the arrival of COVID to the country. Panel (b) documents the monthly quantity of calls to the Police's Family Help Phone line (*Fono Familia*, # 149), which are only available until November 2021. Panel (c) documents monthly unemployment rates as reported by the Central Bank of Chile. And Panel (d) documents relative changes in the amount of time which individuals are estimated to spend in residential areas based on Google's Community Mobility Reports in the country. Vertical red lines document the data of first school closure, and first reopening.



(a) Attendance

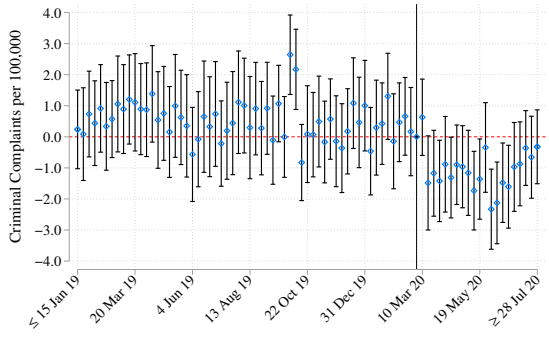
(b) Formal Employment

(c) School Assistants (rate)

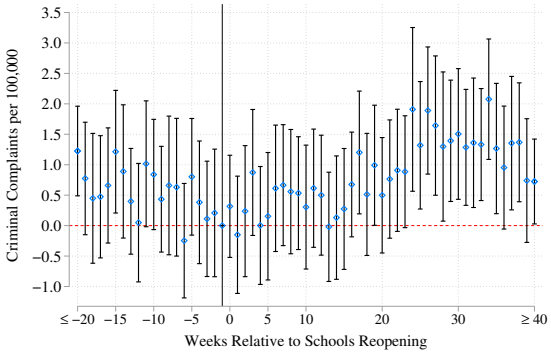
(d) School Assistants (experience)

Figure A14: Correlation Between Mediators: Baseline and Post-Pandemic

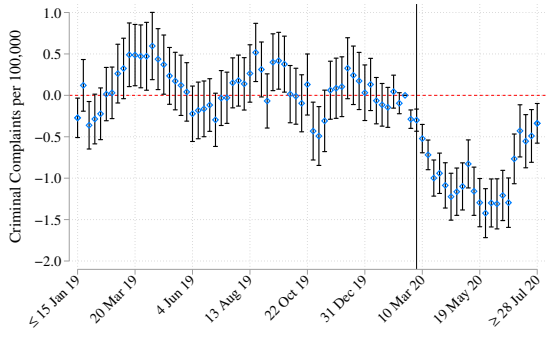
Notes to Figure A14: Scatter plot and linear fit documents the correlation between measures indicated in plot titles in the pre-pandemic period (baseline is 2019), and post-pandemic school reopening period (2021). For monthly measures (panels (a) and (b)) baseline and post-pandemic periods refer to the month of November. In panel (a), given data restrictions, measurement of attendance is slightly different – at baseline it refers to average daily attendance, while post-pandemic it refers to the proportion of individuals attending at least 10 days per month. Each point is a municipality, with sizes indicating the municipal population.



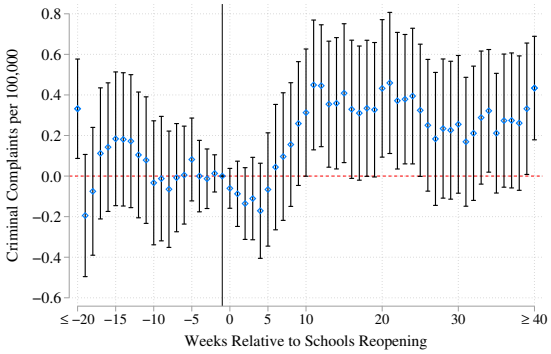
(a) Intra-family violence (closure)



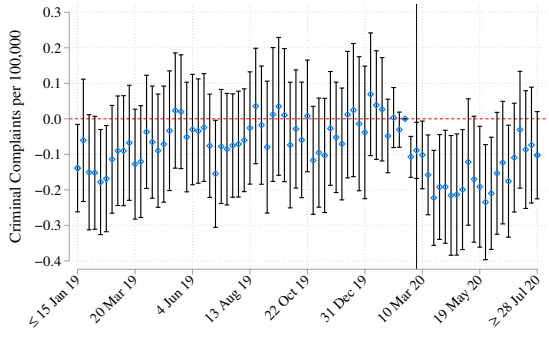
(b) Intra-family violence (opening)



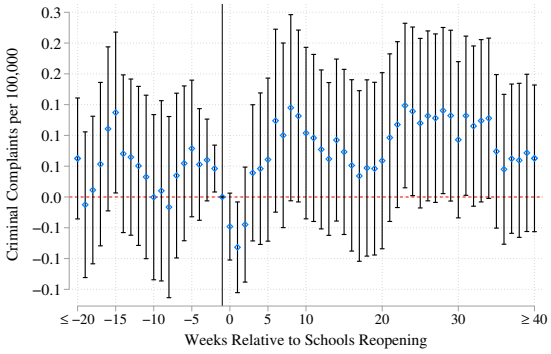
(c) Sexual abuse (closure)



(d) Sexual abuse (opening)



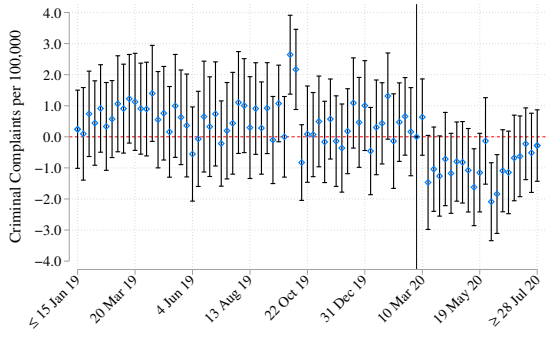
(e) Sexual abuse (closure)



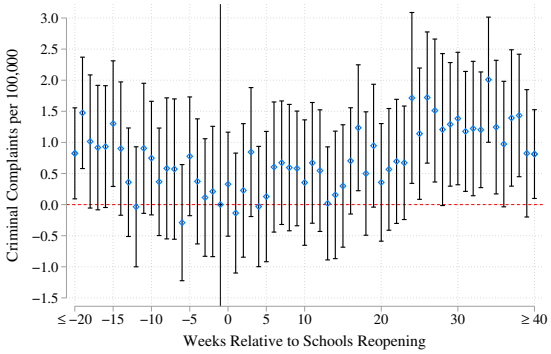
(f) Sexual abuse (opening)

Figure A15: Event Study Estimates of School Closure and Reopening on Reporting of Violence Against Children

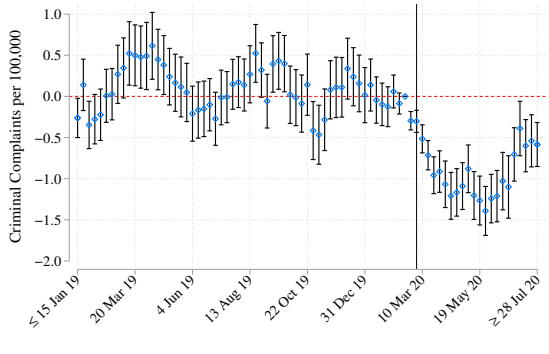
Notes to Figure A15: Event studies are documented as described in equation 2, and details follow those in Notes to Figure 3. Here, lock-down controls are consistently included. Implementations follow Clarke and Schythe (2020). All other details follow those described in equation 2.



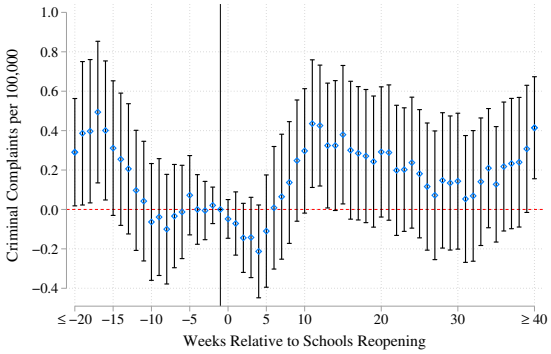
(a) Intra-family violence (closure)



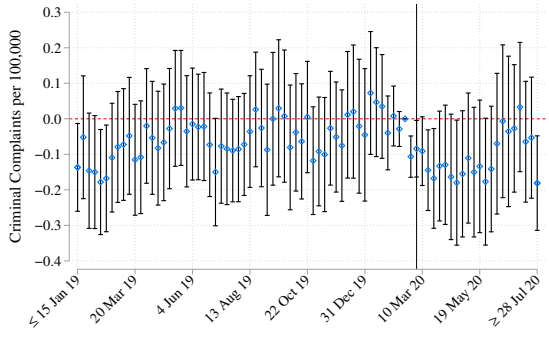
(b) Intra-family violence (opening)



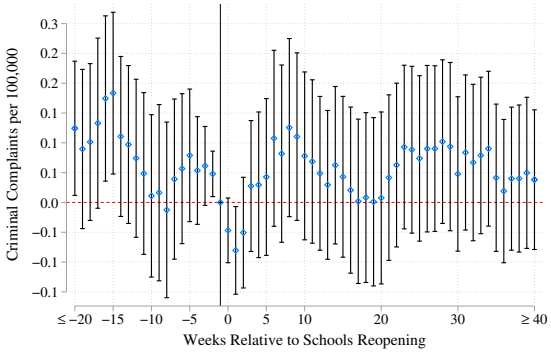
(c) Sexual abuse (closure)



(d) Sexual abuse (opening)



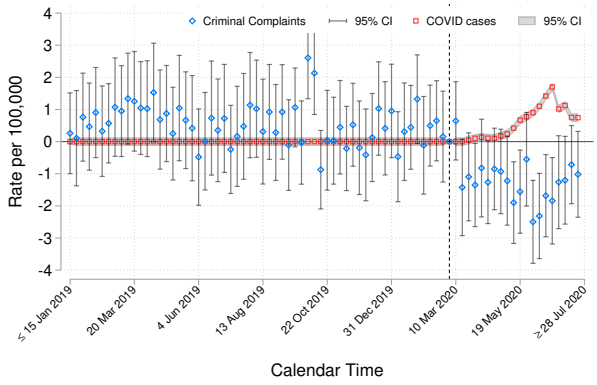
(e) Sexual abuse (closure)



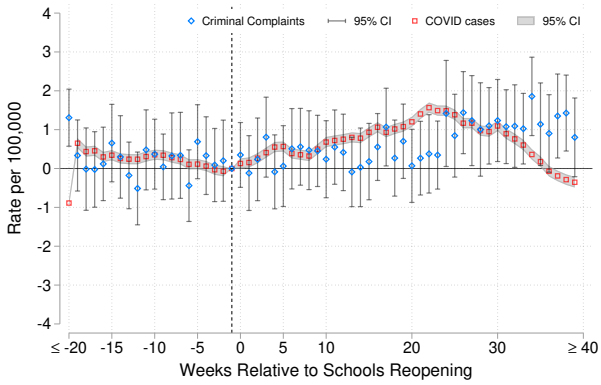
(f) Sexual abuse (opening)

Figure A16: Event Study Estimates of School Closure and Reopening on Reporting of Violence Against Children

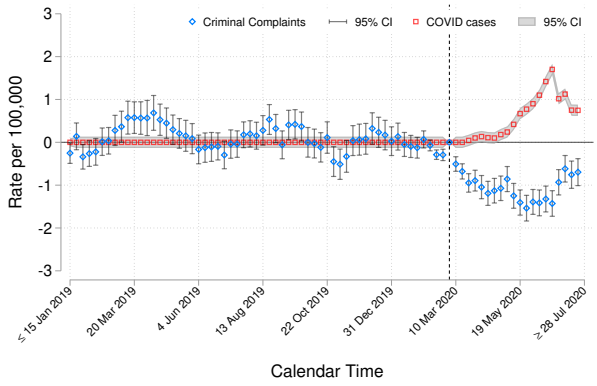
Notes to Figure A16: Event studies are documented as described in equation 2, and details follow those in Notes to Figure 3. Here, lock-down and COVID-intensity controls are consistently included. All other details follow those described in equation 2.



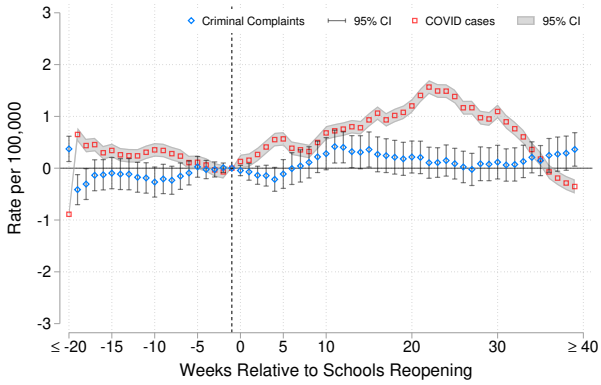
(a) Intra-family Violence (closure)



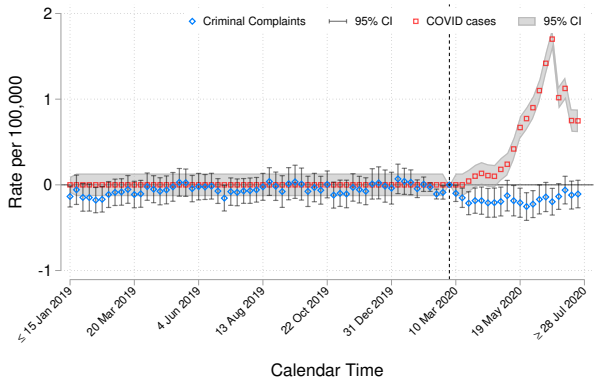
(b) Intra-family Violence (opening)



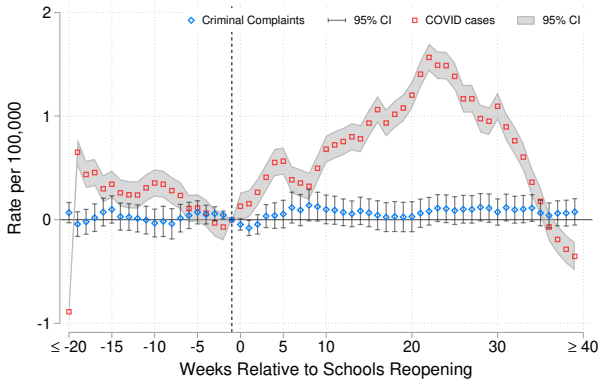
(c) Sexual abuse (closure)



(d) Sexual abuse (opening)



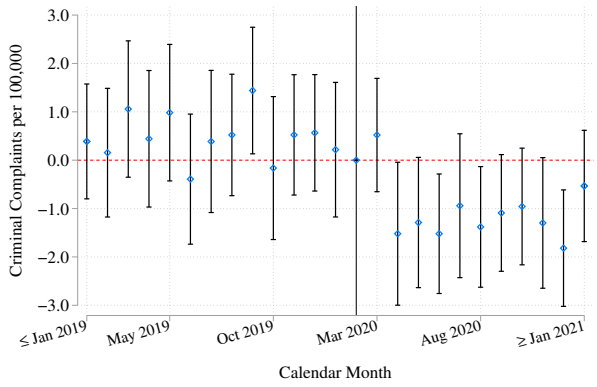
(e) Rape (closure)



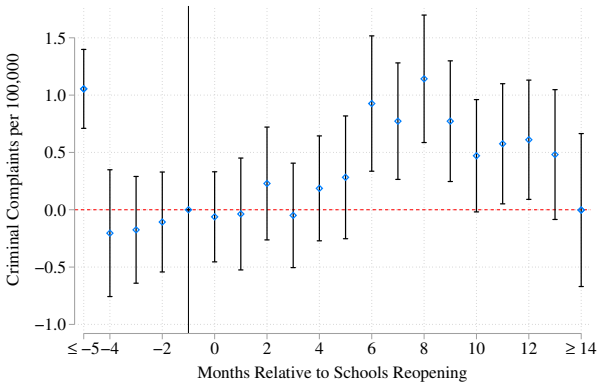
(f) Rape (opening)

Figure A17: Event Studies and Infection Rates

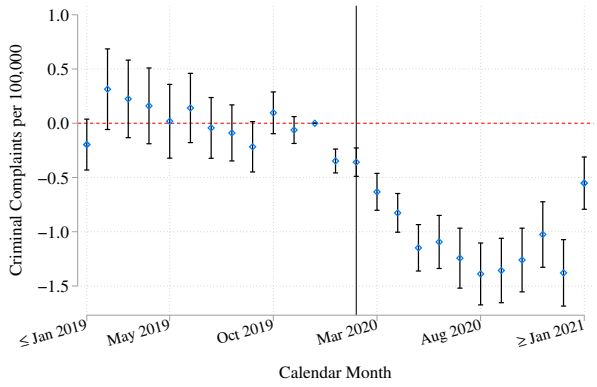
Notes to Figure A17: Event studies are estimated as in Figure 3, however now presenting lags and leads simultaneously: criminal complaints of violence against children (blue points and grey CIs), and recorded COVID-19 cases (red points and grey shaded CIs). Estimates for recorded COVID-19 cases are identical in all panels considering school closure ((a), (c) and (e)), and school all panels considering reopening ((b), (d) and (f)). In each case, identical scaling is used across the left and right plots (school closure and reopening), but scales for each type of violence are altered for ease of visualisation. All other details follow those described in Figure 3.



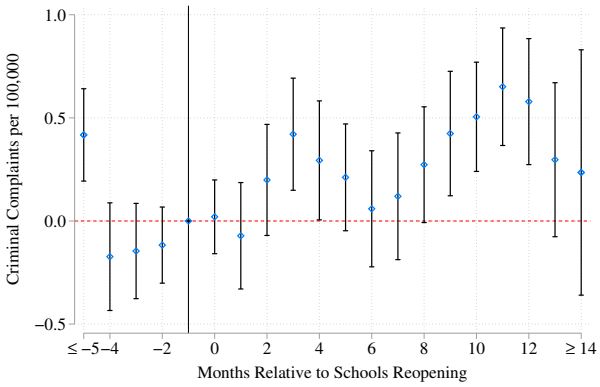
(a) Intra-family Violence (closure)



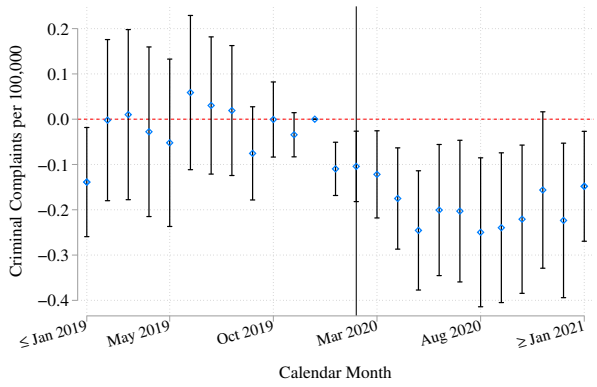
(b) Intra-family Violence (opening)



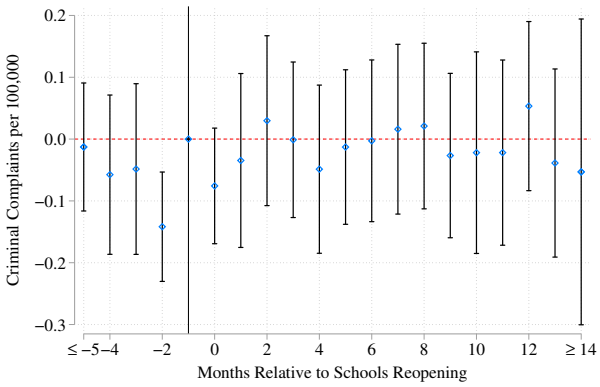
(c) Sexual abuse (closure)



(d) Sexual abuse (opening)



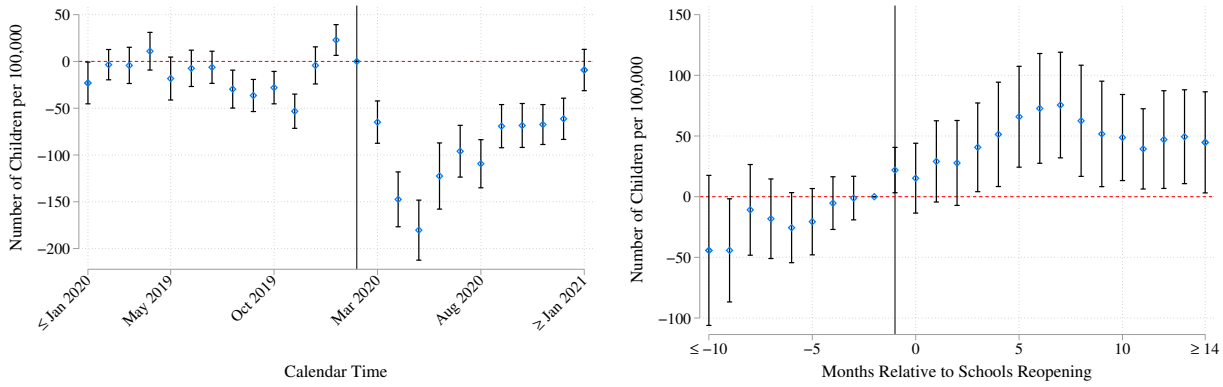
(e) Rape (closure)



(f) Rape (opening)

Figure A18: Event Studies by Month: Rates of Crime Reporting

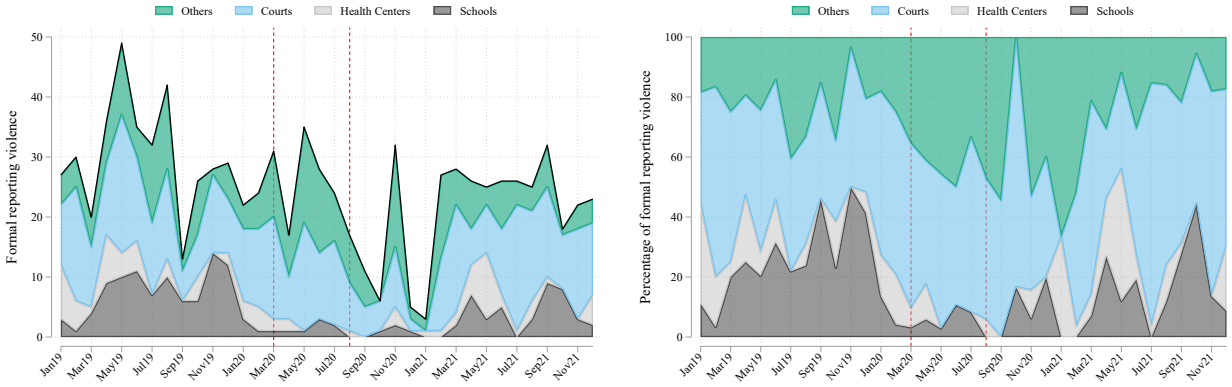
Notes to Figure A18: Event studies follow those in Figure 3, however grouping lags and leads by month (-1 refers to the month of February 2020, a period of school vacations). Point estimates refer to lags and leads in months, and 95% confidence intervals are based on standard errors clustered by municipality. All other details follow those laid out in Notes to Figure 3.



(a) School Closure (b) Reopening

Figure A19: Event Studies by Month: Program Entries to Child Protection System

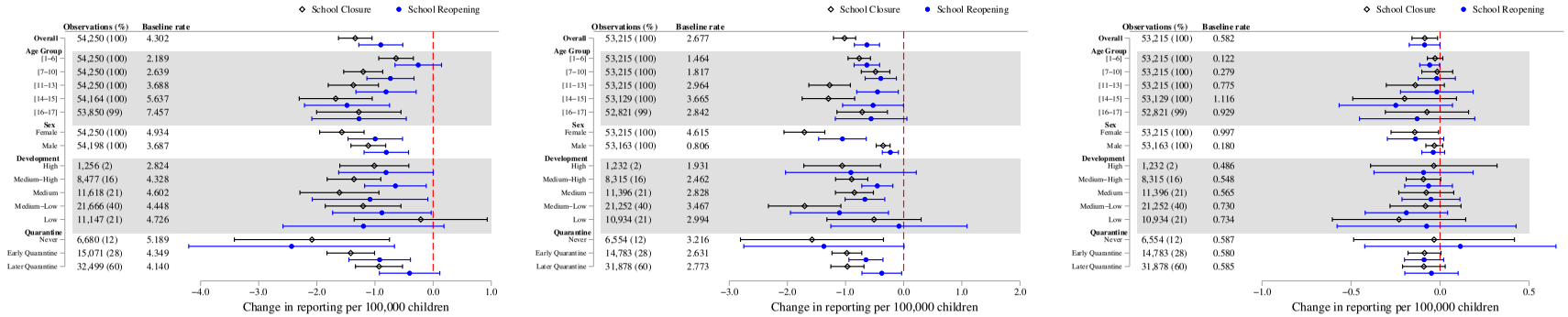
Notes to Figure A19: Monthly event studies are presented based on cases which are entered into the child protection system. These data are only available at the monthly level, and refer to the number of programs in which a child begins participating which is managed by the child protection system. Monthly lags and leads are reported as blue points, alongside 95% confidence intervals based on standard errors clustered by municipality. All municipalities are weighted by the population of 5-18 year-olds in each municipality.



(a) Violence Reporting by Reporting Channel (b) Percentage of Violence Reporting by Reporting Channel

Figure A20: Actual Reporting Channels Reported by a Single Child Protection Office in a Large Municipality

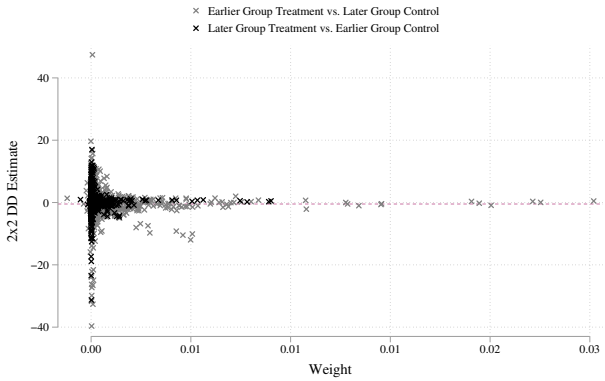
Notes to Figure A20: All reports of violence received by the child protection office of a large municipality in Santiago are displayed. These are broken down by reporting channels as from schools, from municipal health care centres, from courts and from other sources. Total numbers of cases by month in this municipality are displayed in the left-hand panel, and absolute proportions are displayed in the right-hand panel. Vertical dotted lines represent dates of school closure and school reopening in the municipality. January and February correspond to summer break in Chile.



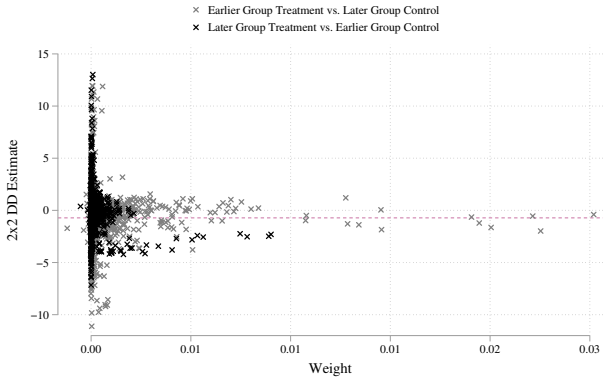
(a) Intra-family Violence (b) Sexual Abuse (c) Rape

Figure A21: Impacts of School Closures and Openings: Demographic and Socio-Economic Variation

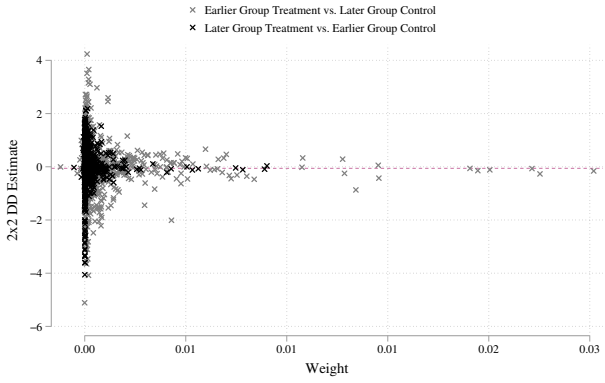
Notes to Figure A21: Estimates (diamonds and circles) and 95% confidence intervals are displayed for models corresponding to sub-groups indicated on the vertical axis. Estimates for each group correspond to coefficients on School Closure (diamonds), and Schools Reopening (circles), following equation 1, with full time-varying controls. In each case, estimates are based on the population or municipality-specific estimation sample, and estimates are consistently weighted by the population of the estimation sample. The total number of municipality by week observations are indicated in "Observations", with the % referring to the percent of the full sample of municipality by week cells. Baseline (pre-2020) rates per 100,000 individuals of each group are displayed as "Baseline rate".



(a) Intra-family Violence



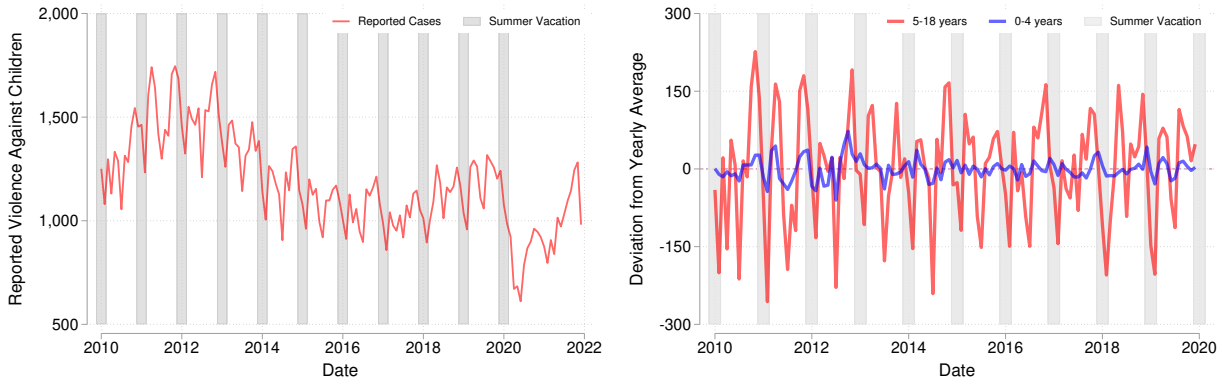
(b) Sexual Abuse



(c) Rape

Figure A22: Weights and 2×2 Double-Difference Estimates in Two-way Fixed Effect Models

Notes to Figure A22: Plots document the double-difference decomposition laid out by (Goodman-Bacon, 2021) to decompose single coefficient estimates on School Reopening displayed in Table 1 of the paper. Here, each cross displays the proportional weight of each municipal×week switching estimate (horizontal axis), as well as the DD estimate for each switching pair (vertical axis), compared with the single-coefficient estimate (dotted horizontal line). Grey crosses represent individual estimates based on the (desired) comparison of treated to not yet treated units, while black crosses represent comparisons of later switchers to earlier switchers.

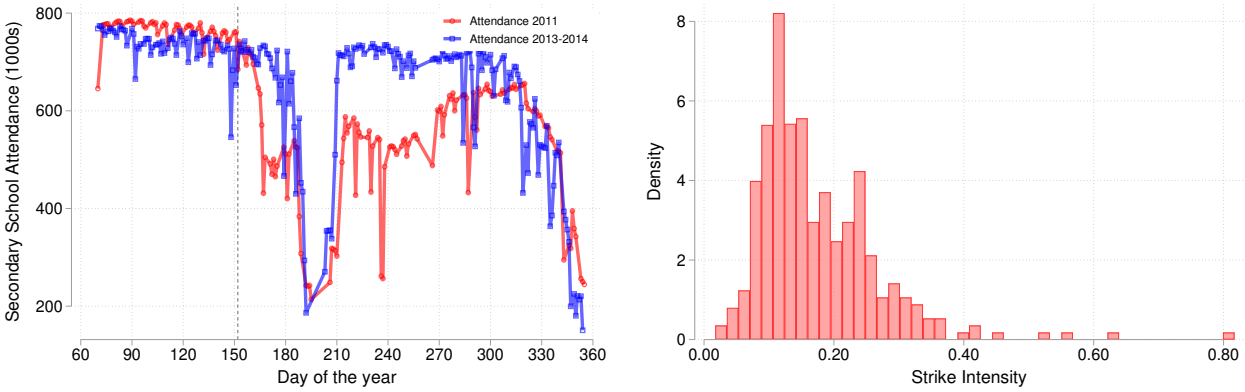


(a) All Cases

(b) Deviation from Yearly Average by Age

Figure A23: Reported Violence Against Children by Month

Notes to Figure A23: Total reported cases of violence against children (intra-family violence, sexual abuse and rape) are plotted over time in panel (a) with shaded areas corresponding to summer vacation months (December-February). Deviations from yearly averages are documented in panel (b) for school-aged (5-17 year olds) and under school-aged (0-4) children.



(a) Attendance Over Time

(b) Strike Intensity

Figure A24: Exposure to Student Strikes

Notes to Figure A24: In panel (a), dashed vertical line indicates 1 June 2011, date where student strikes ramp up (see González, 2020). Points refer to the number of students attending secondary school each day for the strike period (2011, red line) and post-strike period (2013-2014, blue line). 2012 given that monthly attendance was reported for a small number of months. Weekends, public holidays and summer holidays are omitted as these have close to 0 recorded attendance. Panel (b) documents strike intensity at the municipal level as the proportion of total days lost in this municipality between June and December 2011.

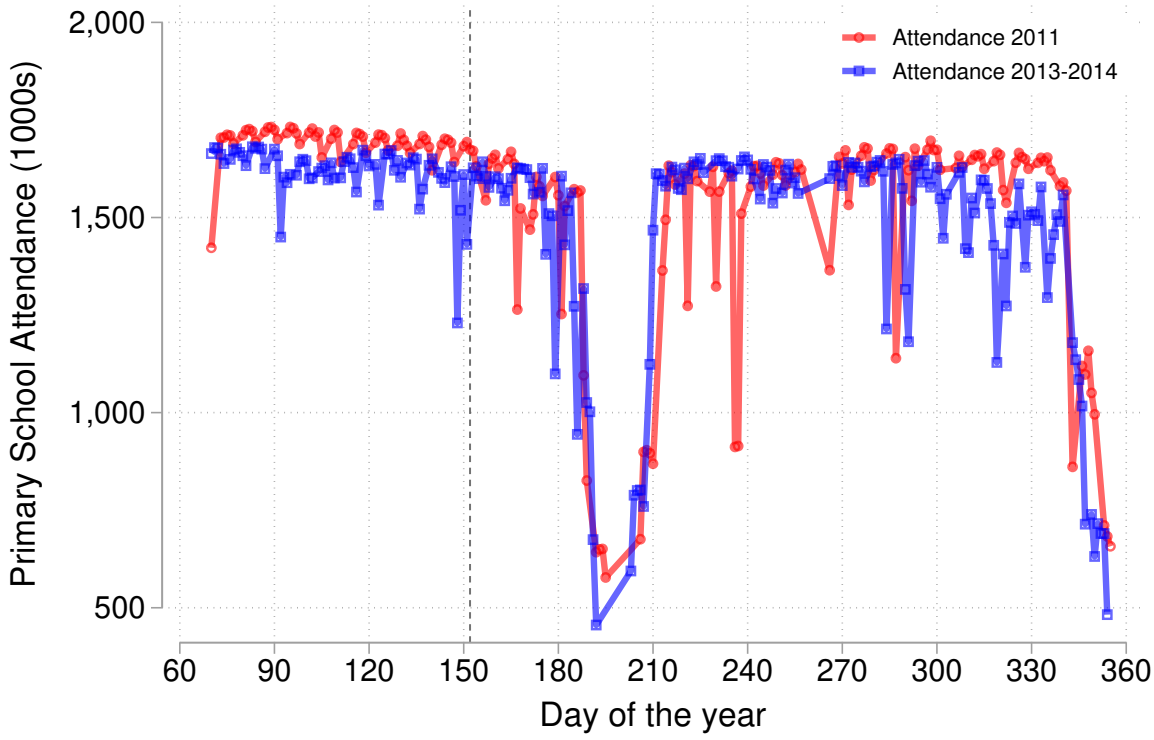


Figure A25: School Strikes and Primary Attendance

Notes to Figure A25: An identical plot to Figure A24 is presented, however here for primary-aged children attendance, instead of secondary-aged children.

A2 Supplementary Tables

Table A1: Summary Statistics of Principal Variables (2019–2021)

	Observations	Mean	Std. Dev.	Min.	Max.
Panel A: Violence Against Children					
Intrafamily Violence	54321	3.93	18.91	0.00	2325.58
Physical Violence (serious)	54321	0.13	2.20	0.00	116.82
Physical Violence (moderate)	54321	2.44	12.62	0.00	1086.96
Psychological Violence	54321	1.35	13.81	0.00	2325.58
Sexual Abuse	53283	2.94	14.09	0.00	1020.41
Rape	53283	0.60	2.18	0.00	80.71
Panel B: Schools					
School Closure	54321	0.32	0.46	0.00	1.00
School Reopening	54250	0.29	0.45	0.00	1.00
Schools Reopening	54250	0.19	0.35	0.00	1.00
Baseline Attendance	54321	0.74	0.11	0.05	0.91
Baseline Educational Specialists	54164	57.62	22.53	15.07	204.11
Composition of Students (private)	54164	0.04	0.12	0.00	1.00
Composition of Students (municipal)	54164	0.54	0.34	0.00	1.00
Composition of Students (priority)	54164	0.66	0.18	0.00	1.00
Panel C: COVID/Other Measures					
Quarantine in place	54321	0.10	0.30	0.00	1.00
COVID cases per 1,000	54321	0.68	1.88	0.00	233.58
PCR tests per 1,000	54321	8.10	10.69	0.00	179.53
PCR test positivity	54321	5.27	9.66	0.00	100.00
COVID-19 vaccinations administered	54321	294.29	1234.20	0.00	34485.00
Proportion of adults in formal employment	54281	0.23	0.08	0.00	0.47
Population between 5 and 18 years	54321	10222.90	15866.86	9.00	127390.00

Notes to Table A1: Summary statistics are displayed across all municipal by week cells for the period of January 2019–December 2021. Panel A documents mean outcomes of violence against children measured as weekly reports per 100,000 children in each municipality. Panel B documents principal measures of school closure and reopening, as well as baseline attendance rates and coverage of school support workers (students per worker). Panel C documents epidemiological, vaccination, and economic controls.

Table A2: Triple Difference versus Conditional Difference-in-differences

	Difference-in-Differences				Triple
	0-4 year	5-10 years	0-4 year	5-10 years	difference
	(1)	(2)	(3)	(4)	(5)
Panel A: 5-10 year olds versus 0-4 year olds					
School Close	-0.630***	-1.189***	-0.439***	-1.069***	-0.630***
	(0.127)	(0.131)	(0.144)	(0.146)	(0.162)
Schools Reopening	-0.085	-0.572***	-0.125	-0.652***	-0.527***
	(0.118)	(0.124)	(0.180)	(0.181)	(0.164)
Observations	54,250	54,250	54,250	54,250	54,250
Coefficient Ratio (Close)				0.899	0.589
Coefficient Ratio (Open)				1.141	0.808
	0-4 year	5-17 years	0-4 year	5-17 years	
Panel B: 5-17 year olds versus 0-4 year olds					
School Close	-0.630***	-1.479***	-0.439***	-1.276***	-0.837***
	(0.127)	(0.111)	(0.144)	(0.133)	(0.163)
Schools Reopening	-0.085	-0.880***	-0.125	-0.924***	-0.799***
	(0.118)	(0.110)	(0.180)	(0.170)	(0.188)
Observations	54,250	54,250	54,250	54,250	54,250
Coefficient Ratio (Close)				0.863	0.656
Coefficient Ratio (Open)				1.049	0.865
Controls			Y	Y	Y

Notes to Table A2: Columns (1) to (4) present difference-in-differences specifications of the impact of school closure and reopening on rates of criminal complaints against children of ages indicated in column headings. Each observation is a municipal by week cell. Columns (1) and (2) are conditional only on time and municipal fixed effects, while (3) and (4) incorporate time-varying controls. Column 5 presents the triple difference specifications comparing changes among older (more exposed) versus younger (less exposed) children. Coefficient ratio refers to the ratio of coefficients on closure or reopening comparing column 4 or column 5 to unconditional difference-in-differences estimates in column 2. Standard errors clustered by municipality are presented in parentheses. *** $p< 0.01$; ** $p< 0.05$; * $p< 0.10$.

Table A3: Actual Violence Reporting Channels Reported by a Single Child Protection Office – Temporal Differences

	Jan-Feb		Mar-Sep		Oct-Dec		Aug-Dec	
	2019	2020	2019	2020	2019	2020	2019	2021
Total Reporting	57	46	227	163	83	43	138	120
Total Reporting by Schools	4	4	57	8	32	4	48	25
Percentage of Reporting by Schools	0.07	0.09	0.25	0.05	0.39	0.09	0.35	0.21
Percentage of Schools Open	–	–	1.00	0.00	1.00	0.08	1.00	0.99

Notes to Tab. A3: Descriptive values document official recorded channels of violence reporting received by a single child protection office in a large municipality in Santiago. Here year by year comparisons are documented of reports received via schools and total reports received in various periods. Jan-Feb is vacations in both years. Mar-Sep covers periods where schools returned in 2019, but were closed in 2020. Oct-Dec covers periods where schools were completely open in 2019, and only very partially open in 2020. Aug-Dec covers periods in which schools sere completely open in 2019, and nearly entirely open in 2021.

Table A4: Modelled Impacts of School Closure and Reopening on Violence Against Children (no January and February)

	Intra-family Violence			Sexual Abuse			Rape		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Binary Reopening Measure									
School Close	-1.582***	-1.124***	-1.132***	-1.010***	-0.961***	-0.961***	-0.103***	-0.062	-0.062
	(0.133)	(0.162)	(0.161)	(0.085)	(0.110)	(0.110)	(0.026)	(0.041)	(0.041)
Schools Reopening	-0.769***	-0.820***	-0.811***	-0.419***	-0.670***	-0.632***	-0.061**	-0.075*	-0.079*
	(0.146)	(0.198)	(0.201)	(0.082)	(0.121)	(0.126)	(0.030)	(0.045)	(0.046)
Test of $\beta = \gamma$ (p-value)	0.000	0.074	0.068	0.000	0.000	0.000	0.144	0.736	0.672
Observations	45,608	45,568	45,499	44,573	44,533	44,464	44,573	44,533	44,464
Baseline Mean	4.979	4.979	4.979	2.540	2.540	2.540	0.611	0.611	0.611
Panel B: Continuous Reopening Measure									
School Close	-1.433***	-0.790***	-0.865***	-0.884***	-0.601***	-0.665***	-0.087***	-0.022	-0.022
	(0.122)	(0.135)	(0.140)	(0.077)	(0.083)	(0.085)	(0.024)	(0.033)	(0.034)
Schools Reopening	-0.656***	-0.405**	-0.524**	-0.178*	-0.000	-0.109	-0.035	-0.002	-0.002
	(0.177)	(0.201)	(0.223)	(0.098)	(0.117)	(0.120)	(0.041)	(0.050)	(0.054)
Test of $\beta = \gamma$ (p-value)	0.000	0.083	0.137	0.000	0.000	0.000	0.193	0.698	0.710
Observations	45,608	45,568	45,499	44,573	44,533	44,464	44,573	44,533	44,464
Baseline Mean	4.979	4.979	4.979	2.540	2.540	2.540	0.611	0.611	0.611
Municipal & WoY FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
Lockdown & Epidemiological controls		Y	Y		Y	Y		Y	Y
School composition controls			Y			Y			Y

Notes to Table A4: Refer to notes to Table 1 of the main text. Identical specifications are estimated, however here consistently removing all observations in months January-February of all years, when schools are closed for summary vacations. *** p< 0.01; ** p< 0.05; * p< 0.10.

Table A5: Impacts of School Closure and Reopening on Sexual Violence (Unadjusted Variables)

	Sexual Abuse			Rape		
	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Binary Reopening Measure						
School Close	-0.972***	-0.945***	-0.946***	-0.098***	-0.086**	-0.087**
	(0.085)	(0.110)	(0.109)	(0.027)	(0.044)	(0.044)
Schools Reopening	-0.384***	-0.617***	-0.613***	-0.049*	-0.070	-0.080*
	(0.072)	(0.112)	(0.118)	(0.028)	(0.046)	(0.046)
Test of $\beta = \gamma$ (p-value)	0.000	0.000	0.000	0.098	0.672	0.860
Observations	54,250	54,210	54,124	54,250	54,210	54,124
Baseline Mean	2.506	2.506	2.506	0.611	0.611	0.611
Panel B: Continuous Reopening Measure						
School Close	-0.886***	-0.642***	-0.685***	-0.094***	-0.063*	-0.060
	(0.082)	(0.092)	(0.095)	(0.026)	(0.036)	(0.038)
Schools Reopening	-0.279***	-0.180	-0.245*	-0.065*	-0.062	-0.056
	(0.095)	(0.124)	(0.126)	(0.039)	(0.049)	(0.053)
Test of $\beta = \gamma$ (p-value)	0.000	0.000	0.000	0.461	0.984	0.936
Observations	54,250	54,210	54,124	54,250	54,210	54,124
Baseline Mean	2.506	2.506	2.506	0.611	0.611	0.611
Municipal & WoY FEs	Y	Y	Y	Y	Y	Y
Lockdown & epidemiological controls		Y	Y		Y	Y
School composition controls			Y			Y

Notes to Table A5: Refer to notes to Table 1 of the main text. Identical specifications are estimated for models where the outcome is sexual abuse or rape, however using original un-smoothed data, where over-reporting occurs on the first day of each month, rather than smoothed data re-assigning excess reporting uniformly across the month. All other details follow those in Table 1. Column numbers (4)-(9) are used here for sake of comparison with column numbers in Table 1. *** p< 0.01; ** p< 0.05; * p< 0.10.

Table A6: Impacts of School Closure and Reopening on Reporting of Violence Against Children by Type of Intra-family Violence

	Physical Violence (Serious)			Physical Violence (Moderate)			Psychological Violence		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Binary Reopening Measure									
School Close	-0.030**	0.002	0.002	-1.148***	-0.796***	-0.796***	-0.415***	-0.370***	-0.374***
	(0.013)	(0.020)	(0.020)	(0.087)	(0.106)	(0.106)	(0.067)	(0.082)	(0.082)
Schools Reopening	-0.023	0.012	0.020	-0.803***	-0.736***	-0.722***	-0.026	-0.153	-0.151
	(0.014)	(0.018)	(0.019)	(0.076)	(0.111)	(0.115)	(0.083)	(0.103)	(0.106)
Test of $\beta = \gamma$ (p-value)	0.681	0.612	0.379	0.000	0.521	0.453	0.000	0.010	0.012
Observations	54,250	54,210	54,124	54,250	54,210	54,124	54,250	54,210	54,124
Baseline Mean	0.189	0.189	0.189	3.191	3.191	3.191	1.649	1.649	1.649
Panel B: Continuous Reopening Measure									
School Close	-0.025*	-0.000	-0.006	-1.019***	-0.488***	-0.561***	-0.386***	-0.264***	-0.266***
	(0.013)	(0.017)	(0.017)	(0.082)	(0.086)	(0.092)	(0.059)	(0.064)	(0.067)
Schools Reopening	-0.017	0.016	0.007	-0.790***	-0.418***	-0.535***	0.075	0.076	0.088
	(0.022)	(0.024)	(0.024)	(0.105)	(0.126)	(0.131)	(0.101)	(0.109)	(0.123)
Test of $\beta = \gamma$ (p-value)	0.733	0.522	0.612	0.029	0.598	0.841	0.000	0.004	0.004
Observations	54,250	54,210	54,124	54,250	54,210	54,124	54,250	54,210	54,124
Baseline Mean	0.189	0.189	0.189	3.191	3.191	3.191	1.649	1.649	1.649
Municipal & WoY FEs	Y	Y	Y	Y	Y	Y	Y	Y	Y
Lockdown & Epidemiological controls		Y	Y		Y	Y		Y	Y
School composition controls			Y			Y			Y

Notes to Table A6: Refer to notes to Table 1 of the main text. Identical specifications are estimated, however here rather than considering total reports of intra-family violence against children (as reported in columns (1)-(3) of Table 1), reports of intra-family violence against children for each sub-class of intra-family violence are reported. These classifications are generated from police reports, and each case of intra-family violence from Table 1 is classified as one (and only one) of the three classes displayed here. All other details follow those in Table 1. *** p< 0.01; ** p< 0.05; * p< 0.10.

Table A7: Weights in Time-Varying Adoption Two-Way Fixed Effect Estimate on School Reopening

Comparison Group	Weights	Average DD Estimate
Panel A: Intra-family violence		
Earlier Treatment vs. Later Control	0.788	-0.637
Later Treatment vs. Earlier Control	0.212	0.127
Panel B: Sexual abuse		
Earlier Treatment vs. Later Control	0.788	-0.544
Later Treatment vs. Earlier Control	0.212	-1.368
Panel C: Rape		
Earlier Treatment vs. Later Control	0.788	-0.074
Later Treatment vs. Earlier Control	0.212	-0.001

Notes to Tab A7: Aggregate values for two-way FE weights are reported (following (Goodman-Bacon, 2021; Goodman-Bacon et al., 2019)) for estimates of impacts of time-varying school closure on outcomes indicated in each panel. These are aggregate values across all points documented in Figure A22. Weights refer to the total proportion of estimates based on each comparison type (earlier treatment vs. later adopter, or later adopter vs. earlier treated), and "Average DD Estimate" refers to the average difference between these groups in a 2×2 DD setting. For all models, the proportion of negative weights following (de Chaisemartin and D'Haultfœuille, 2020) is 0.

Table A8: Published School Vacation Dates from Government Resolutions

Region	2011		2012		2013		2014		2015		2016		2017		2018		2019	
	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End
Region I																		
Summer	07/03	23/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	26/12	05/03	26/12
Winter	11/07	22/07	09/07	20/07	08/07	19/07	07/07	18/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	08/07	19/07
Region II																		
Summer			01/03	21/12	04/03	20/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter			17/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	08/07	19/07
Region III																		
Summer	03/03	16/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	18/12	03/03	14/12	06/03	22/12	05/03	21/12	05/03	18/12
Winter	11/07	22/07	17/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	17/07	28/07	16/07	27/07	15/07	26/07
Region IV																		
Summer	03/03	16/12	05/03	21/12	04/03	20/12	03/03	19/12	04/03	18/12	03/03	21/12	06/03	22/12	05/03	19/12	05/03	20/12
Winter	18/07	29/07	16/07	27/07	15/07	26/07	14/07	25/07	13/07	26/07	11/07	24/07	10/07	23/07	09/07	22/07	13/07	27/07
Region V																		
Summer	03/03	21/12	05/03	21/12	05/03		05/03		04/03		03/03	16/12	05/03		05/03		05/03	20/12
Winter	11/07	22/07	09/07	20/07	08/07	19/07	07/07	18/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	15/07	26/07
Region VI																		
Summer	03/03	16/12	05/03	21/12	05/03	24/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	18/07	29/07	09/07	20/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	08/07	19/07
Region VII																		
Summer	03/03	16/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	15/12	05/03	21/12	05/03	20/12
Winter	18/07	29/07	16/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	08/07	19/07
Region VIII																		
Summer	03/03	21/12	05/03	19/12	05/03	19/12	06/03	18/12	04/03	17/12	03/03	12/12	06/03	08/12	05/03	21/12	05/03	19/12
Winter	18/07	30/07	17/07	28/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	17/07	27/07	15/07	26/07
Region IX																		
Summer	03/03	16/12	05/03	21/12	05/03	20/12	05/03	23/12	04/03	22/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	18/07	29/07	16/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	17/07	28/07	16/07	27/07	15/07	26/07
Region X																		
Summer	03/03	16/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	18/07	29/07	16/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	18/07	29/07	17/07	28/07	17/07	27/07	15/07	26/07
Region XI																		
Summer	03/03	23/12	05/03	28/12			05/03	26/12	04/03	24/12	03/03	23/12	06/03	29/12	05/03	28/12	05/03	31/12
Winter	11/07	29/07	09/07	27/07			14/07	01/08	13/07	31/07	11/07	29/07	10/07	28/07	09/07	27/07	08/07	26/07

Notes to Table A8: Dates are hand-coded from resolutions issued by the Ministry of Education which define vacation dates for each of the 16 regions of Chile. Region XVI (Ñuble) was created in 2018 and previously belonged to the Region VII (Biobío), and as such prior to 2018 simply takes the same dates as those indicated in Region VII. A small number of dates could not be determined from Ministerial documents.

Table A8 (Cont'd): Published School Vacation Dates from Government Resolutions

Region	2011		2012		2013		2014		2015		2016		2017		2018		2019	
	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End	Start	End
Region XII																		
Summer	03/03	23/12	05/03	23/12	26/03	27/12	05/03	12/12	04/03	18/12	03/03	23/12	06/03	29/12	05/03	28/12	05/03	27/12
Winter	11/07	29/07	09/07	27/07	08/07	26/07	14/07	01/08	13/07	31/07	11/07	29/07	10/07	28/07	09/07	27/07	08/07	26/07
Region XIII																		
Summer	03/03	21/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	19/12	03/03	21/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	11/07	22/07	09/07	20/07	08/07	19/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	17/07	27/07	15/07	26/07
Region XIV																		
Summer	03/03	23/12	05/03	21/12	05/03	20/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	18/07	29/07	16/07	27/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	22/07	09/07	20/07	08/07	19/07
Region XV																		
Summer	03/03	16/12	05/03	21/12	05/03	24/12	05/03	19/12	04/03	18/12	03/03	16/12	06/03	22/12	05/03	21/12	05/03	20/12
Winter	11/07	22/07	09/07	20/07	08/07	29/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	09/07	20/07	08/07	19/07
Region XVI																		
Summer	03/03	21/12	05/03	19/12	05/03	19/12	06/03	18/12	04/03	17/12	03/03	12/12	06/03	08/12	05/03	21/12	05/03	19/12
Winter	18/07	30/07	17/07	28/07	15/07	26/07	14/07	25/07	13/07	24/07	11/07	22/07	10/07	21/07	17/07	27/07	15/07	26/07

See notes on previous page.

Table A9: Summary Statistics of Principal Variables (School Strikes and Vacation Analysis)

	Observations	Mean	Std. Dev.	Min.	Max.
Panel A: School Vacation Analysis					
All Violence Against Children	126636	1.03	3.85	0.00	571.43
Attendance	126636	0.61	0.37	0.00	1.00
Vacation Period	126636	0.26	0.44	0.00	1.00
Term Time (non Post-vacations)	126636	0.70	0.46	0.00	1.00
Post-vacation Period	126636	0.02	0.15	0.00	1.00
Population	126636	10192.29	15665.85	6.00	129878.00
Panel B: School Strikes Analysis					
All Violence	31356	55.75	102.40	0.00	5882.35
Intrafamily Violence	31356	40.93	89.10	0.00	5882.35
Sexual Abuse	31356	11.12	40.80	0.00	1587.30
Rape	31356	3.70	24.12	0.00	1538.46
Strike Period	31356	0.09	0.29	0.00	1.00
Strike Intensity	31356	0.15	0.10	0.00	0.82
Population of Secondary Students	31356	3261.07	4855.77	1.00	40619.00

Notes to Table A9: Summary statistics are displayed across all municipal by week cells for the period of January 2012–December 2017 in the case of vacation analysis (Panel A). Statistics are displayed across all municipal by month cells for the period of January 2010–December 2017 in the case of analysis of school strikes (Panel B).

Table A10: School Strikes in 2011 and Intra-Family Violence Against Children (Triple Difference)

	All Complaints (1)	Intra-family Violence (2)	Sexual Abuse (3)	Rape (4)
Strike Intensity × Strike	-43.39***	-25.11**	-5.828	-12.45***
	(15.35)	(10.87)	(5.536)	(4.252)
Observations	31,356	31,356	31,356	31,356
R-squared	0.08	0.09	0.06	0.06
Scaled Effect	-4.05	-2.34	-0.54	-1.16
Mean Dep. Var.	55.75	40.93	11.12	3.70

Notes to Table A10: Estimates are presented following details laid out in (6) and Notes to Table 4, however here considering as an outcome differences in reporting rates between secondary and primary children, implementing a triple difference specification, where the third difference is schooling level (secondary is exposed, primary is not). All other details follow those in Table 4. *** p< 0.01; ** p< 0.05; * p< 0.10.

A3 Considering Counterfactual Projections and Evolution of Violence Rates

In this Appendix we describe an activity which seeks to compared observed rates of violence with rates of violence based off of "counterfactual" projections assuming that recent trends are informative of what may have happened in the absence of the pandemic. Based off of such counterfactual projections, we can consider the differences between actual and projected reporting rates, and additionally, what proportion of these differences can be explained by the school closure and reopening channels.

This counterfactual activity, which we describe below, assumes that we can infer projections based on levels and trends in reporting prior to the arrival of COVID-19 and associated school closures. However, a broad stream of literature (Bullinger et al., 2020; Bhalotra et al., 2024; Erten et al., 2022; Pereda and Díaz-Faes, 2020) suggests that violence may have in fact increased, given the change in risk factors associated with child maltreatment (i.e., time spent at home, and increase in economic strain, financial hardship, mental stress, and family conflict). If this was the case, the counterfactuals may actually under-estimate the true expected reporting if rates of reporting had been maintained constant. Thus, we also conduct additional tests, allowing for potential increases in rates of violence. Overall, we see these counterfactual projections as a bounding exercise of the range of reports that may have been missed due to school closures and slow recovery, under different assumptions of changes in true rates of violence.

A3.1 Methods

Using data on reporting incidence per 100,000 individuals aged under 18 for each of the three outcomes discussed above, we estimate counterfactual trends from observable (pre-COVID) data as:

$$\widehat{\text{Reporting}}_{mt}^{post} = \widehat{\alpha}^{pre} + \widehat{\mu}_{WoY}^{pre} + \widehat{\phi}_{m}^{pre}, \tag{A1}$$

where projected reporting in the post school closure and reopening period in municipality m and week t, denoted $\widehat{\text{Reporting}}_{mt}^{post}$ is estimated by projecting pre-closure averages in each municipality and week of year. In Clarke et al. (2022), we document an extended methodology which also allowed for temporal reporting trends. Estimated week of year fixed effects $\widehat{\mu}_{WoY}^{pre}$ allow us to capture cyclical (within year) variation, while flexible time trends allow us to capture secular changes in reporting over time. We discuss the selection of prediction periods and modeling of secular trends below. A key factor of this counterfactual projection is that it estimates all coefficients and fixed effects entirely off pre-COVID data, allowing for the projection of such trends into the post-COVID period, abstracting from the actual effects of COVID and school closure on violence reporting.

Based on real and projected trends, we calculate differences between real and 'expected' reporting rates in the absence of COVID and school closures. This is simply:

$$\text{Difference}_{mt}^{post} = \text{Reporting}_{mt}^{post} - \widehat{\text{Reporting}}_{mt}^{post} \tag{A2}$$

Note that here, we can calculate a difference *for each* municipality and week. In the following results, we consider total differences in reporting measured as *absolute case* differentials at a national level in each time period t by converting these per capita reporting differentials into total reporting differentials, and aggregating over the entire country. This is:

$$\text{Reporting Differential}_t = \sum_{m=1}^{346} \text{Difference}_{mt}^{post} \times \frac{Population_{mt}}{100,000}. \tag{A3}$$

As we calculate the Reporting Differential at each period t, we can observe in a dynamic way how this evolves, considering both pre- and post-school reopening periods.

In a second step, we re-estimate counterfactual reporting now controlling additionally for the channel of school closures and reopenings. This is calculated as:

$$\widehat{\text{Reporting}}_{mt}^{post}\Big|_{SO} = \widehat{\alpha}^{pre} + \widehat{\mu}_{WoY}^{pre} + \widehat{\phi}_{m}^{pre} + \widehat{\delta}\text{School Opening}_{mt}, \tag{A4}$$

where all details follow equation A1, but we additionally control for School Opening$_{mt}$ (SO), which takes the value of 1 while schools are fully open, 0 while schools are fully closed, and then the proportion of students whose school is reopen upon school reopening periods. We follow identical procedures in calculating reporting differentials from equation A3, however now conditional on School Opening channels, and document relative movements when calculating the unconditional counterfactual in equation A1 and the conditional counterfactual in equation A4 to estimate the proportional contribution of school closures and reopenings to reporting differentials.

Finally, to account for potential increase in violence, Reporting Differentials were calculated considering increases by fixed rates, for example by 10%, as below:

$$\text{Difference}_{mt}^{post,\Delta 10\%} = \text{Reporting}_{mt}^{post} - \left(\widehat{\text{Reporting}}_{mt}^{post}\right) \times 1.10. \tag{A5}$$

We display Reporting Differentials following equation A1 under these alternative sensitivity assumptions.

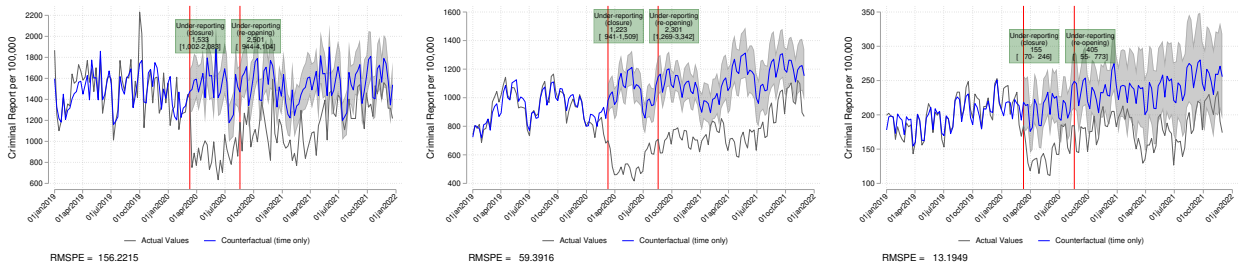
Finally, in conducting inference on these projections, we must take account of the fact that counterfactual outcomes are estimated based on observed data, and hence are subject to sampling uncertainty inherent in this estimation procedure. To conduct inference, we undertake a block bootstrap procedure, resampling over Chile's 346 municipalities to maintain the time-series dependence within each municipality.

A3.2 Results

Figure A26 presents the counterfactual projections. In panel A we document simple projections: how would complaints of violence against children perform if simply projecting cyclical (week of year) and municipal-specific factors forward, estimated off the pre-pandemic period. In each case, we observe that such projections perform well in predicting in-sample (2019) and out-of-sample (2020), up until the week of school closure. We then observe sharp declines when comparing actual complaints (thin grey line), to counterfactual predictions (thick blue line). We observe that over time, these lines nearly converge, with actual reporting nearly reaching counterfactual projections, though this convergence is slow, only approaching predicted levels by around the fourth quarter of 2021, over a year after the first schools were reopened. Indeed, when mapping estimated differences between real and projected complaints, we estimate that in the school closure period 1,533 (95% CI: 1,002–2,083) cases of intra-family violence against children were not reported, 1,223 (95% CI: 941–1,509) cases of sexual abuse against children were not reported, and 155 (95% CI: 70–246) cases of rape against children were not reported. Somewhat similar values are observed in the post-reopening period, but these are estimated over a much longer period.

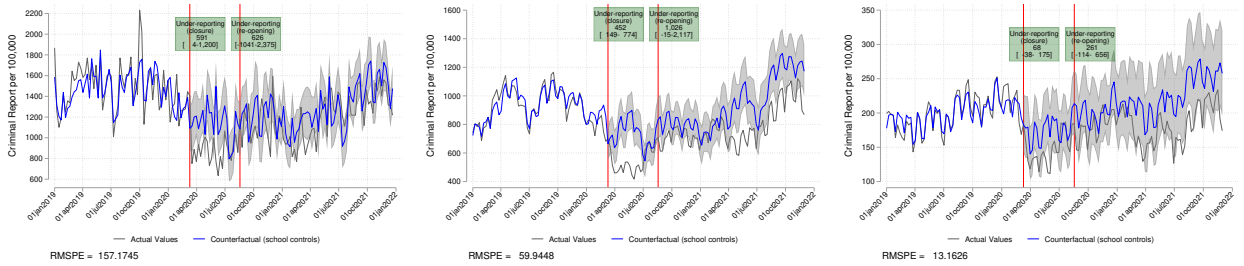
In Panel B, we consider alternative projections, now 'turning off' the school reporting channel. If school closure accounts for the full difference in reporting, we would expect that these counterfactual

Panel A: Simple Counterfactual (Time Only)



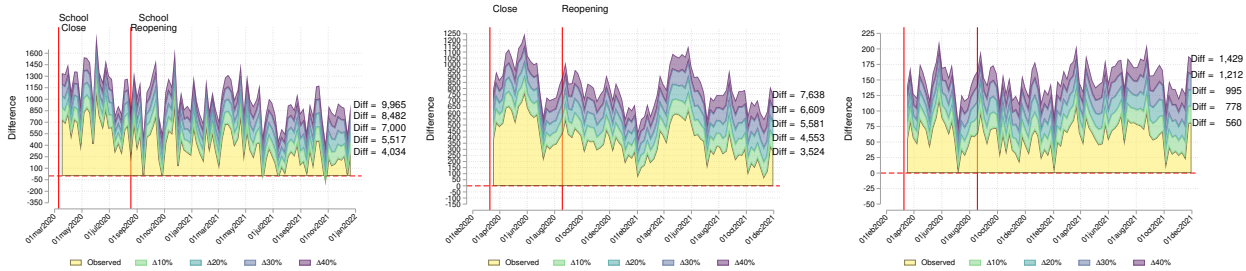
(a) Intra-family Violence (b) Sexual Abuse (c) Rape

Panel B: Counterfactual (No School Channel)



(d) Intra-family Violence (e) Sexual Abuse (f) Rape

Panel C: Projected Under-reporting



(g) Intra-family Violence (h) Sexual Abuse (i) Rape

Figure A26: Reporting, Projected Reporting, and Under-reporting Under Various Counterfactual Assumptions

Notes to Figure A26: Subfigures (a)-(c) document actual reporting (grey line) and projected counterfactuals (blue line), with 95% bootstrap confidence intervals for (a) intra-family violence, (b) sexual abuse, and (c) rape against children, following equation A1. Counterfactuals are estimated using optimal temporal trends (linear in subfigure (a), and quadratic in (b) and (c), and pre-pandemic prediction periods, with root mean squared prediction errors displayed in the bottom left corner of each panel. Subfigures (d)-(f) document identical counterfactual procedures, but now 'switching off' the school closure and reopening channel, following equation A4. Aggregate differences between real and projected reporting for the closure and reopening period along with bootstrapped 95% CIs are displayed in green squares, and week by week reporting differentials are displayed in subfigures (g)-(i), along with sensitivity testing following equation A5, in which rates of projected violence are allowed to increase in the post-pandemic period.

and observed trends would now entirely overlap. While we observe large movements, we do not observe that school closure can explain away *the entirety* of under-reported cases in these projections. Comparing panel (a) to panel (d), we observe that school closure can explain away 934 of the 1,533

estimated 'missing' intra-family violence reports during the school closure period, and that partial school closure can account for 1,848 of the 2,501 'missing' intra-family violence reports during the school opening period. In the case of sexual abuse and rape, we observe similar patterns, with school closure explaining between 41% (rape) to 57% (sexual abuse) of the drop in reporting observed over the entire period. These results are broadly similar if additionally controlling for COVID case rates, testing and positivity rates, as well as municipality lockdown status.

Finally, if, as prior literature suggest (see Section 4.3), violence increases during this time, the models presented in Panels A and B could be read as the lower bound of reports missing due to school closure. In panel C, we consider alternative projections where rather than assuming that historical and cyclical trends predict counterfactual (no COVID-19 related school closures) outcomes, we assume that violence may actually have increased. As we do not know the magnitude of the increase,[28] we document the difference between counterfactuals and actual reporting under alternative assumptions of increases in underlying violence by {10, 20, 30, 40}%. The results suggest that, if true rates of violence had actually increased by 10% above trend, rather than 4,034 unreported cases of intra-family violence against children in aggregate, this would rise to 5,517, with broadly similar proportional changes in the case of sexual abuse (from 3,524 to 4,553) and rape (from 560 to 778).

[28] A survey conducted with US parents in the first weeks of the pandemic documents that 20% of the parents report hitting or spanking their child in the past 2 weeks, with 5% reporting doing so more often than usual, while 1 in 4 recognizing an increase in conflict during that time (Lee et al., 2021). Similarly estimates accounting for factors associated with violence point to increases of up to 29% in referrals of violence against children in a specific US-setting (Prettyman, 2021b), while an increase of around 10% in the use of women's shelters in Chile was observed as a consequence of lockdowns (Bhalotra et al., 2024), pointing to an increases in violence within the household based on objective measures. a wide range of values ranging from 0 to as much as a 40% increase.

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